Britannia Industries Ltd: Ratio Analysis Problem

Solution: Step-by-Step Working

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Basis: average of 31 March 2025 and 31 March 2026 balances for FY2026 where the ratio uses an average; year-end balances for FY2025, because the 31 March 2024 balance sheet is not available. All amounts are in ₹ crore unless stated. Money is rounded to two decimals, ratios to two decimals, percentages to two decimals and days to one decimal; full figures are carried through the working.

Step 0: Derived variables used in the ratios

Derived variable	Variables included	Calculation
Cost of goods sold	Cost of materials consumed + Purchases of stock-in-trade + Changes in inventories + Conversion charges	FY2026: 9,045.68 + 2,028.01 + (−21.42) + 764.69 = 11,816.96 FY2025: 8,608.64 + 1,993.16 + (−67.96) + 802.20 = 11,336.04
Purchases	Purchases of raw and packing materials + Purchases of stock-in-trade	FY2026: 9,152.52 + 2,028.01 = 11,180.53 FY2025: 8,576.31 + 1,993.16 = 10,569.47
EBIT	Profit before tax + Finance costs (interest)	FY2026: 3,293.34 + 110.80 = 3,404.14 FY2025: 2,867.77 + 137.10 = 3,004.87
Long-term debt	Non-current bank borrowings + Current maturities of bank term loans + Lease liabilities (non-current + current)	FY2026: 285.83 + 426.67 + 1.62 (0.36 + 1.26) = 714.12 FY2025: 712.50 + 191.67 + 1.13 (0.48 + 0.65) = 905.30
Total capital	Long-term debt + Total equity	FY2026: 714.12 + 4,641.23 = 5,355.35 FY2025: 905.30 + 3,886.55 = 4,791.85
Net fixed assets	Property, plant and equipment + Capital work-in-progress	FY2026: 2,426.37 + 38.35 = 2,464.72 FY2025: 2,509.41 + 81.97 = 2,591.38
Total trade payables	Dues of micro and small enterprises + Dues of other creditors	FY2026: 20.40 + 1,761.47 = 1,781.87 FY2025: 27.18 + 1,565.22 = 1,592.40
Tangible net worth	Total equity − Intangible assets	FY2026: 4,641.23 − 3.39 = 4,637.84 FY2025: 3,886.55 − 5.96 = 3,880.59
Market value of equity	Market price per share (₹) × Number of shares outstanding (24.0868296 crore)	FY2026: 5,334.16 × 24.0868296 = 128,483.00 FY2025: 4,793.21 × 24.0868296 = 115,453.23

Liquidity

1. Current ratio

Formula: Current assets ÷ Current liabilities

Year	Working	Result
FY2026	4,187.68 ÷ 3,940.22	**1.06**
FY2025	3,436.43 ÷ 3,348.13	**1.03**

Interpretation: Current assets cover current liabilities 1.06 times against 1.03; the cushion is thin but better, which is typical of a fast-moving consumer goods business that is financed by suppliers.

2. Quick (acid-test) ratio

Formula: (Cash and cash equivalents + Bank balances + Current investments + Trade receivables) ÷ Current liabilities

Year	Working	Result
FY2026	(20.79 + 70.03 + 1,617.02 + 404.74) ÷ 3,940.22 = 2,112.58 ÷ 3,940.22	**0.54**
FY2025	(38.19 + 67.45 + 1,071.12 + 379.63) ÷ 3,348.13 = 1,556.39 ÷ 3,348.13	**0.46**

Interpretation: Liquid assets cover 54 paise of every rupee of current liabilities against 46 paise; the improvement comes from higher current investments, and the ratio is below 1 only because inventories are excluded.

3. Working capital (₹ crore)

Formula: Current assets − Current liabilities

Year	Working	Result
FY2026	4,187.68 − 3,940.22	**247.46**
FY2025	3,436.43 − 3,348.13	**88.30**

Interpretation: Working capital rose by ₹159.16 crore to ₹247.46 crore, so the short-term financial buffer is nearly three times its earlier level.

4. Cash ratio

Formula: (Cash and cash equivalents + Bank balances + Current investments) ÷ Current liabilities

Year	Working	Result
FY2026	(20.79 + 70.03 + 1,617.02) ÷ 3,940.22 = 1,707.84 ÷ 3,940.22	**0.43**
FY2025	(38.19 + 67.45 + 1,071.12) ÷ 3,348.13 = 1,176.76 ÷ 3,348.13	**0.35**

Interpretation: Cash and liquid investments alone cover 43% of current liabilities against 35%, so Britannia can meet a large part of its short-term obligations without relying on receivables or inventory.

Solvency

5. Times interest earned

Formula: EBIT ÷ Interest

Year	Working	Result
FY2026	3,404.14 ÷ 110.80	**30.72 times**
FY2025	3,004.87 ÷ 137.10	**21.92 times**

Interpretation: Earnings before interest and tax cover interest 30.72 times against 21.92 times, because profit rose while finance costs fell; the debt burden is very light.

6. Times fixed charges earned

Formula: (EBIT + Rent) ÷ (Interest + Rent)

Year	Working	Result
FY2026	(3,404.14 + 77.63) ÷ (110.80 + 77.63) = 3,481.77 ÷ 188.43	**18.48 times**
FY2025	(3,004.87 + 90.28) ÷ (137.10 + 90.28) = 3,095.15 ÷ 227.38	**13.61 times**

Interpretation: Even after counting rent as a fixed charge, earnings cover fixed charges 18.48 times against 13.61 times, so the commitment burden is comfortably small.

7. Debt to equity

Formula: Long-term debt ÷ Total equity

Year	Working	Result
FY2026	714.12 ÷ 4,641.23	**0.15**
FY2025	905.30 ÷ 3,886.55	**0.23**

Interpretation: Long-term debt is 15 paise for every rupee of equity against 23 paise, because debt fell from ₹905.30 crore to ₹714.12 crore while equity grew; borrowing plays a small and shrinking part in the capital structure.

8. Debt to total assets

Formula: Long-term debt ÷ Total assets

Year	Working	Result
FY2026	714.12 ÷ 8,940.44	**0.08**
FY2025	905.30 ÷ 8,019.72	**0.11**

Interpretation: Long-term debt funds about 8 paise of every rupee of assets against 11 paise, so the asset base is financed overwhelmingly by equity and operating liabilities rather than long-term borrowing.

Funds management

9. Receivables to sales and average collection period

Formula: Average receivables ÷ Sales; then × 365

Year	Working	Result
FY2026	Average receivables = (404.74 + 379.63) ÷ 2 = 392.19; 392.19 ÷ 18,445.82 = 0.0213; × 365 = 7.8 days	**2.13%; 7.8 days**
FY2025	Year-end receivables = 379.63; 379.63 ÷ 17,295.92 = 0.0219; × 365 = 8.0 days	**2.19%; 8.0 days**

Interpretation: Receivables are only about 2% of sales, and customers pay in roughly 8 days in both years, which reflects a largely cash-and-short-credit distribution model.

10. Inventory turnover and days in inventory

Formula: Cost of goods sold ÷ Average inventory; then 365 ÷ turnover

Year	Working	Result
FY2026	Average inventory = (1,219.92 + 1,128.90) ÷ 2 = 1,174.41; 11,816.96 ÷ 1,174.41 = 10.06 times; 365 ÷ 10.0620 = 36.3 days	**10.06 times; 36.3 days**
FY2025	Year-end inventory = 1,128.90; 11,336.04 ÷ 1,128.90 = 10.04 times; 365 ÷ 10.0417 = 36.3 days	**10.04 times; 36.3 days**

Interpretation: Inventory turns about ten times a year, or every 36 days, in both years; this is steady and suits a business that sells short-shelf-life bakery products.

11. Payables period

Formula: (Year-end trade payables ÷ Purchases) × 365

Year	Working	Result
FY2026	(1,781.87 ÷ 11,180.53) × 365 = 0.1594 × 365	**58.2 days**
FY2025	(1,592.40 ÷ 10,569.47) × 365 = 0.1507 × 365	**55.0 days**

Interpretation: Britannia takes about 58 days to pay suppliers against 55 days, which is longer credit from suppliers; the figure is overstated because payables also include freight, advertising and conversion charges.

12. Cash conversion cycle

Formula: Days in inventory + Collection period − Payables period

Year	Working	Result
FY2026	36.27 + 7.76 − 58.17	**−14.1 days**
FY2025	36.35 + 8.01 − 54.99	**−10.6 days**

Interpretation: The cycle is negative, so suppliers fund the operating cycle; the negative gap widened by about 3.5 days because payables stretched while inventory and collection days held.

13. Total asset turnover

Formula: Sales ÷ Average total assets

Year	Working	Result
FY2026	Average total assets = (8,940.44 + 8,019.72) ÷ 2 = 8,480.08; 18,445.82 ÷ 8,480.08	**2.18 times**
FY2025	Year-end total assets = 8,019.72; 17,295.92 ÷ 8,019.72	**2.16 times**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Each rupee of assets generates about ₹2.18 of sales against ₹2.16, a near-flat result; the figure is held down by roughly ₹3,700 crore of investments that earn treasury income rather than sales.

14. Fixed asset turnover

Formula: Sales ÷ Average net fixed assets

Year	Working	Result
FY2026	Average fixed assets = (2,464.72 + 2,591.38) ÷ 2 = 2,528.05; 18,445.82 ÷ 2,528.05	**7.30 times**
FY2025	Year-end fixed assets = 2,591.38; 17,295.92 ÷ 2,591.38	**6.67 times**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Sales per rupee of plant rose from 6.67 to 7.30 times because sales grew while net fixed assets fell, so existing capacity is being worked harder.

15. Capital (investment) turnover

Formula: Sales ÷ Average total capital

Year	Working	Result
FY2026	Average capital = (5,355.35 + 4,791.85) ÷ 2 = 5,073.60; 18,445.82 ÷ 5,073.60	**3.64 times**
FY2025	Year-end capital = 4,791.85; 17,295.92 ÷ 4,791.85	**3.61 times**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Long-term capital generates about ₹3.64 of sales per rupee against ₹3.61, so capital productivity is high and stable.

Profitability

16. Gross margin

Formula: (Sales − Cost of goods sold) ÷ Sales

Year	Working	Result
FY2026	(18,445.82 − 11,816.96) ÷ 18,445.82 = 6,628.86 ÷ 18,445.82	**35.94%**

Year	Working	Result
FY2025	(17,295.92 − 11,336.04) ÷ 17,295.92 = 5,959.88 ÷ 17,295.92	**34.46%**

Interpretation: Gross margin widened by about 1.5 percentage points to 35.94%, because cost of goods sold grew more slowly than sales, which shows pricing and cost control.

17. EBIT (operating) margin

Formula: EBIT ÷ Sales

Year	Working	Result
FY2026	3,404.14 ÷ 18,445.82	**18.45%**
FY2025	3,004.87 ÷ 17,295.92	**17.37%**

Interpretation: Operating margin improved by about 1.1 percentage points to 18.45%, even though advertising and promotion rose sharply; this margin includes other income.

18. Net margin

Formula: Net income ÷ Sales

Year	Working	Result
FY2026	2,561.72 ÷ 18,445.82	**13.89%**
FY2025	2,130.72 ÷ 17,295.92	**12.32%**

Interpretation: Net income per rupee of sales rose to 13.89% from 12.32%, helped by better margins, lower interest and a lower effective tax rate.

19. Return on assets

Formula: Net income ÷ Average total assets

Year	Working	Result
FY2026	2,561.72 ÷ 8,480.08	**30.21%**
FY2025	2,130.72 ÷ 8,019.72 (year-end assets)	**26.57%**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Assets earned 30.21% against 26.57%, driven by the higher net margin since asset turnover barely changed; the FY2025 figure uses year-end assets.

20. Return on total capital

Formula: Net income ÷ Average total capital

Year	Working	Result
FY2026	2,561.72 ÷ 5,073.60	**50.49%**
FY2025	2,130.72 ÷ 4,791.85 (year-end capital)	**44.47%**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Long-term capital earned about 50.5% against 44.5%, a very high return that reflects strong margins on a modest capital base.

21. Return on equity

Formula: Net income ÷ Average equity

Year	Working	Result
FY2026	Average equity = (4,641.23 + 3,886.55) ÷ 2 = 4,263.89; 2,561.72 ÷ 4,263.89	**60.08%**
FY2025	2,130.72 ÷ 3,886.55 (year-end equity)	**54.82%**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Shareholders earned 60.08% against 54.82%, an exceptional return that rose on higher margins and a lower tax burden even though leverage fell.

22. Return on tangible net worth

Formula: Net income ÷ Average (Equity − Intangible assets)

Year	Working	Result
FY2026	Tangible net worth: 4,637.84 and 3,880.59; average = 4,259.22; 2,561.72 ÷ 4,259.22	**60.15%**
FY2025	2,130.72 ÷ 3,880.59 (year-end)	**54.91%**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: Because intangibles are tiny (₹3.39 crore), the return on tangible net worth is almost identical to ROE, which shows that earnings are not supported by acquired goodwill.

Linking ratio

23. Extended Du Pont

Formula: Pretax margin × Asset turnover × Total leverage × Tax retention = ROE

Year	Working	Result
FY2026	Pretax margin = 3,293.34 ÷ 18,445.82 = 17.85%; asset turnover = 2.1752; leverage = 8,480.08 ÷ 4,263.89 = 1.9888; tax retention = 2,561.72 ÷ 3,293.34 = 0.7778; product = 0.1785 × 2.1752 × 1.9888 × 0.7778 = 60.08% (equals ROE of 60.08%)	**17.85% × 2.18 × 1.99 × 0.78 = 60.08%**
FY2025	Pretax margin = 2,867.77 ÷ 17,295.92 = 16.58%; asset turnover = 2.1567; leverage = 8,019.72 ÷ 3,886.55 = 2.0635; tax retention = 2,130.72 ÷ 2,867.77 = 0.7430; product = 0.1658 × 2.1567 × 2.0635 × 0.7430 = 54.82% (equals ROE of 54.82%)	**16.58% × 2.16 × 2.06 × 0.74 = 54.82%**

Note: For FY2025 the balance sheet of 31 March 2024 is not available, so year-end balances are used.

Interpretation: The ROE gain comes from a better pretax margin and higher tax retention; turnover is flat and leverage fell, so profitability rather than borrowing drives the return.

Shareholder and market ratios

24. Basic earnings per share (₹)

Formula: (Net income − Preferred dividends) ÷ Weighted average shares

Year	Working	Result
FY2026	(2,561.72 − 0) ÷ 24.0868296 crore shares	**₹ 106.35**
FY2025	(2,130.72 − 0) ÷ 24.0868296 crore shares	**₹ 88.46**

Interpretation: Earnings per share grew 20.2% to ₹106.35, as profit rose while the share count stayed unchanged.

25. Book value per share (₹)

Formula: Total equity ÷ Shares outstanding

Year	Working	Result
FY2026	4,641.23 ÷ 24.0868296	**₹ 192.69**
FY2025	3,886.55 ÷ 24.0868296	**₹ 161.36**

Interpretation: Net assets per share rose to ₹192.69 from ₹161.36 as retained profit built up after dividends.

26. Price-earnings ratio

Formula: Market price per share ÷ EPS

Year	Working	Result
FY2026	5,334.16 ÷ 106.3536	**50.15 times**
FY2025	4,793.21 ÷ 88.4600	**54.19 times**

Interpretation: Investors pay about 50 times earnings against 54 times; the multiple eased because earnings grew faster than the share price, but it still signals high growth and quality expectations.

27. Capitalization rate (earnings yield)

Formula: EPS ÷ Market price per share

Year	Working	Result
FY2026	106.3536 ÷ 5,334.16	**1.99%**
FY2025	88.4600 ÷ 4,793.21	**1.85%**

Interpretation: The market capitalizes earnings at about 2.0% against 1.8%, the reciprocal of the P/E, which is low and implies expectations of strong growth.

28. Price to book value

Formula: Market price per share ÷ Book value per share

Year	Working	Result
FY2026	5,334.16 ÷ 192.6875	**27.68 times**
FY2025	4,793.21 ÷ 161.3558	**29.71 times**

Interpretation: The market values equity at about 27.7 times book value against 29.7 times, a very high premium justified by a 60% return on equity.

29. Dividend yield

Formula: Dividend per share paid ÷ Market price per share

Year	Working	Result
FY2026	75.00 ÷ 5,334.16	**1.41%**
FY2025	73.50 ÷ 4,793.21	**1.53%**

Interpretation: Cash dividends give a yield of 1.41% against 1.53%; the dividend per share rose, but the share price rose faster, so most of the return to investors comes from price gains.

30. Dividend payout ratio

Formula: Cash dividends ÷ Net income

Year	Working	Result
FY2026	1,806.51 ÷ 2,561.72	**70.52%**
FY2025	1,770.38 ÷ 2,130.72	**83.09%**

Interpretation: Payout dropped from 83.09% to 70.52% of profit, so Britannia kept more earnings for the business while still distributing a large share.

31. Sustainable growth rate

Formula: ROE on beginning equity × (1 − Payout ratio)

Year	Working	Result
FY2026	(2,561.72 ÷ 3,886.55) × (1 − 0.7052) = 0.6591 × 0.2948	**19.43%**
FY2025	(2,130.72 ÷ 3,527.52) × (1 − 0.8309) = 0.6040 × 0.1691	**10.22%**

Interpretation: Earnings can grow about 19.4% a year from internal funds against 10.2%, because a higher return on beginning equity combines with a larger retention rate.

Failure prediction

32. Altman Z-score

Formula: 1.2 T1 + 1.4 T2 + 3.3 T3 + 0.6 T4 + 0.999 T5

Step 1: the five terms, with the variables each one uses (year-end balances, FY2026 and FY2025).

Term	Ratio	Variables included	FY2026 calculation	FY2025 calculation
T1	Working capital ÷ Total assets	Working capital = Current assets − Current liabilities	(4,187.68 − 3,940.22) ÷ 8,940.44 = 247.46 ÷ 8,940.44 = 0.0277	(3,436.43 − 3,348.13) ÷ 8,019.72 = 88.30 ÷ 8,019.72 = 0.0110
T2	Retained earnings ÷ Total assets	Retained earnings (Note 19); Total assets	4,367.77 ÷ 8,940.44 = 0.4885	3,613.09 ÷ 8,019.72 = 0.4505

Term	Ratio	Variables included	FY2026 calculation	FY2025 calculation
T3	EBIT ÷ Total assets	EBIT = Profit before tax + Finance costs; Total assets	3,404.14 ÷ 8,940.44 = 0.3808	3,004.87 ÷ 8,019.72 = 0.3747
T4	Market value of equity ÷ Total liabilities	Market value of equity = Market price per share × Shares outstanding; Total liabilities	128,483.00 ÷ 4,299.21 = 29.8853	115,453.23 ÷ 4,133.17 = 27.9333
T5	Sales ÷ Total assets	Sales = Revenue from operations; Total assets	18,445.82 ÷ 8,940.44 = 2.0632	17,295.92 ÷ 8,019.72 = 2.1567

Step 2: apply the weights and add.

Term	Weight	FY2026: weight × value	FY2025: weight × value
T1	1.2	1.2 × 0.0277 = 0.0332	1.2 × 0.0110 = 0.0132
T2	1.4	1.4 × 0.4885 = 0.6840	1.4 × 0.4505 = 0.6307
T3	3.3	3.3 × 0.3808 = 1.2565	3.3 × 0.3747 = 1.2365
T4	0.6	0.6 × 29.8853 = 17.9312	0.6 × 27.9333 = 16.7600
T5	0.999	0.999 × 2.0632 = 2.0611	0.999 × 2.1567 = 2.1545
Z	**Sum**	**21.97**	**20.79**

Result: FY2026 Z = 21.97 and FY2025 Z = 20.79. Both exceed 2.99, so both years fall in the safe zone (above 2.99 safe; 1.80 to 2.99 grey; below 1.80 distress).

Interpretation: Both scores are far above the 2.99 safe threshold (about 22.0 against 20.8), so the model signals negligible bankruptcy risk; the very large market value of equity drives much of the score.

The extended Du Pont product in ratio 23 equals the return on equity in ratio 21 for both years, which confirms the calculations.